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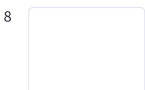
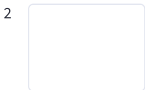
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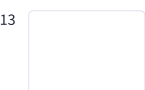
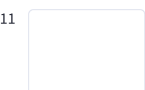
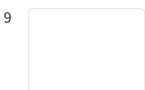
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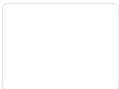
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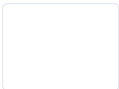
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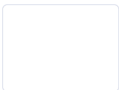
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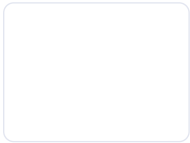
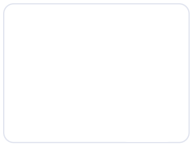
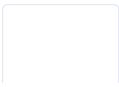
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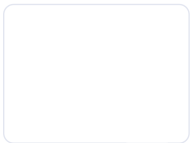
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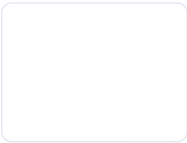
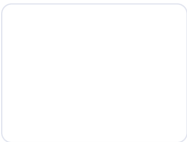


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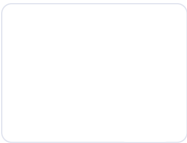
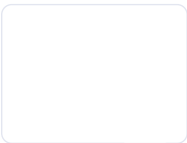




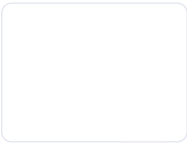
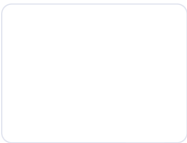
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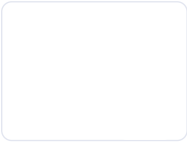
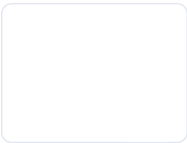
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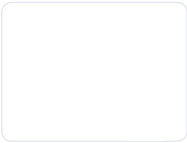
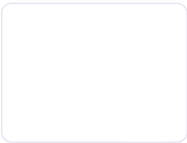
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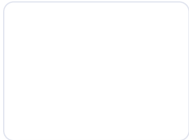
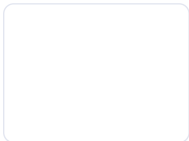
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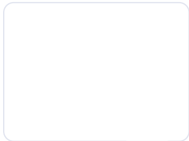
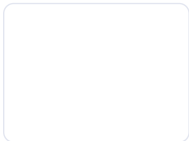
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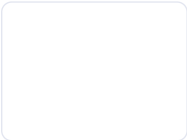


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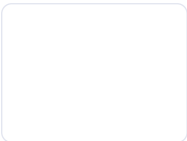
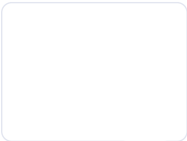


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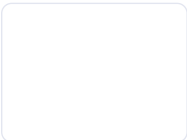
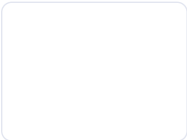




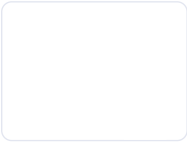
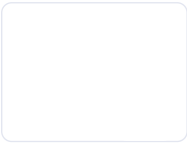
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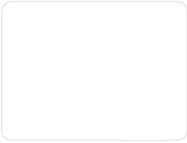
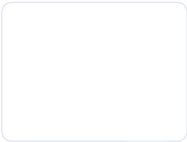
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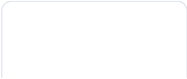
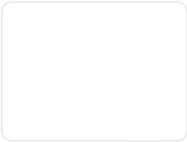
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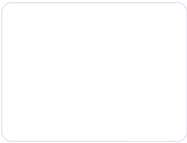
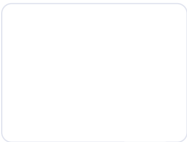


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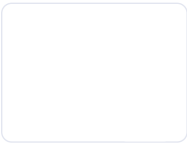
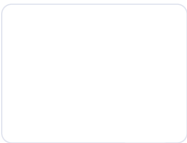




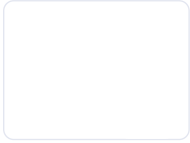
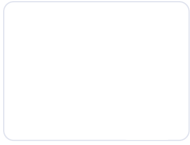
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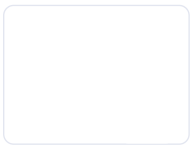
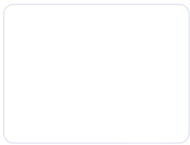
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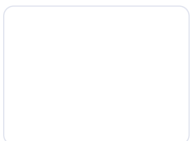
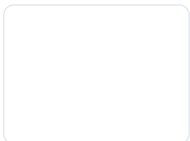
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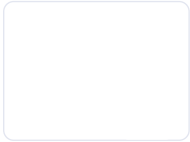
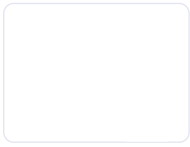
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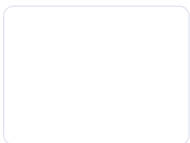
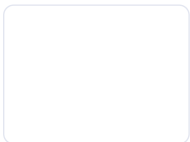
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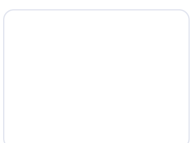
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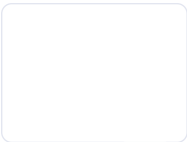


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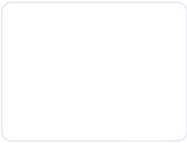
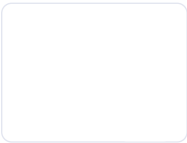


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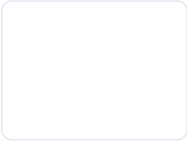
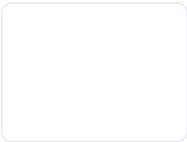




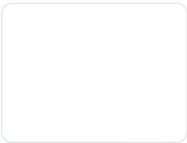
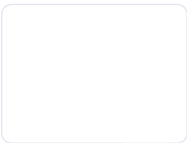
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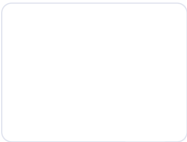
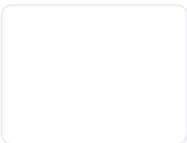
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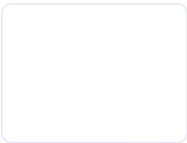
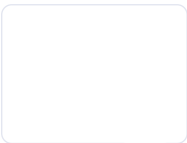
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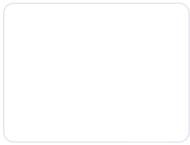
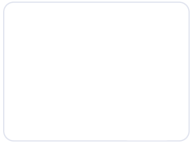
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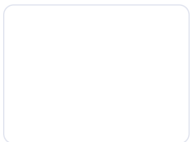
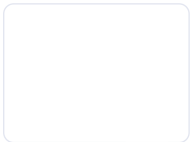
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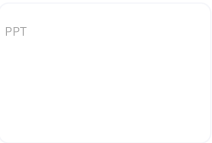
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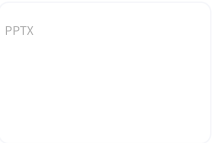
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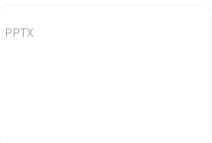
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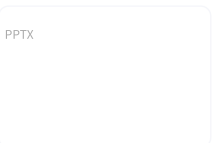
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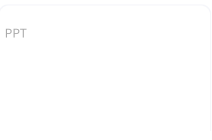
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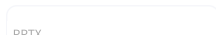
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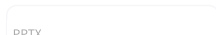
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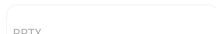
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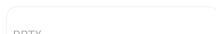
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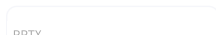
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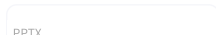
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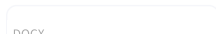
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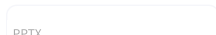
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Met Extraction iron, copper, cobalt etc.

1. **EXTRACTION OF METALS A guide** for A level students KNOCKHARDY PUBLISHING

2. **INTRODUCTION This Powerpoint show** is one of several produced to help students understand selected topics at AS and A2 level Chemistry. It is based on the requirements of the AQA and OCR specifications but is suitable for other examination boards. Individual students may use the material at home for revision purposes or it may be used for classroom teaching if an interactive white board is available. Accompanying notes on this, and the full range of AS and A2 topics, are available from the KNOCKHARDY SCIENCE WEBSITE at...

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KNOCKHARDY PUBLISHING EXTRACTION OF METALS

3. **CONTENTS • Theory of** extraction • Extraction of iron • Conversion of iron into steel • Extraction of aluminium • Extraction of titanium • Extraction of chromium • Extraction of sodium • Recycling EXTRACTION OF METALS
4. **Before you start** it would be helpful to... • Recall the layout of the reactivity series • Recall definitions of reduction, oxidation and redox EXTRACTION OF METALS
5. **GENERAL PRINCIPLES OCCURRENCE • ores** of some metals are very common (iron, aluminium) • others occur only in limited quantities in selected areas • high grade ores are cheaper to process because, ores need to be purified before being reduced to the metal
6. **GENERAL PRINCIPLES THEORY The method** used to extract metals depends on the... • purity required • energy requirements • cost of the reducing agent • position of the metal in the reactivity series
7. **GENERAL PRINCIPLES REACTIVITY SERIES K** Na Ca Mg Al C Zn Fe H Cu Ag • lists metals in descending reactivity • hydrogen and carbon are often added • the more reactive a metal the less likely it will be found in its pure, or native, state • consequently, it will be harder to convert it back to the metal.
8. **GENERAL PRINCIPLES METHODS** - GENERAL Low in series occur native or Cu, Ag extracted by roasting an ore Middle of series metals below carbon are extracted by reduction Zn, Fe of the oxide with carbon or carbon monoxide High in series reactive metals are extracted using electrolysis Na, Al - an expensive method due to energy costs Variations can occur due to special properties of the metal.
9. **GENERAL PRINCIPLES METHODS** - SPECIFIC • reduction of metal oxides with carbon IRON • reduction of metal halides with a metal TITANIUM • reduction of metal oxides by electrolysis ALUMINIUM • reduction of metal oxides with a metal CHROMIUM
10. **IRON**
11. **EXTRACTION OF IRON GENERAL PROCESS** • occurs in the BLAST FURNACE • high temperature process • continuous • iron ores are REDUCED by carbon / carbon monoxide • is possible because iron is below carbon in the reactivity series
12. **RAW MATERIALS HAEMATITE** - Fe₂O₃ a source of iron COKE fuel / reducing agent CHEAP AND PLENTIFUL LIMESTONE conversion of silica into slag (calcium silicate) – USED IN THE CONSTRUCTION INDUSTRY AIR source of oxygen for combustion EXTRACTION OF IRON
13. **THE BLAST FURNACE IN** THE BLAST FURNACE IRON ORE IS REDUCED TO IRON. THE REACTION IS POSSIBLE BECAUSE CARBON IS ABOVE IRON IN THE REACTIVITY SERIES Click on the letters to see what is taking place A B B C D E F G
14. **THE BLAST FURNACE COKE**, LIMESTONE AND IRON ORE ARE ADDED AT THE TOP A Now move the cursor away from the tower
15. **THE BLAST FURNACE HOT AIR** IS BLOWN IN NEAR THE BOTTOM OXYGEN IN THE AIR REACTS WITH CARBON IN THE COKE. THE REACTION IS HIGHLY EXOTHERMIC AND GIVES OUT HEAT. B B CARBON + OXYGEN CARBON + HEAT DIOXIDE C + O₂ CO₂ Now move the cursor away from the tower
16. **THE BLAST FURNACE THE** CARBON DIOXIDE PRODUCED REACTS WITH MORE CARBON TO PRODUCE CARBON MONOXIDE CARBON + CARBON CARBON DIOXIDE MONOXIDE C + CO₂ 2CO C Now move the cursor away from the tower
17. **THE BLAST FURNACE THE** CARBON MONOXIDE REDUCES THE IRON OXIDE D CARBON + IRON CARBON + IRON MONOXIDE OXIDE DIOXIDE 3CO + Fe₂O₃ 3CO₂ + 2Fe REDUCTION INVOLVES REMOVING OXYGEN Now move the cursor away from the tower
18. **THE BLAST FURNACE SILICA** IN THE IRON ORE IS REMOVED BY REACTING WITH LIME PRODUCED FROM THE THERMAL DECOMPOSITION OF LIMESTONE CALCIUM SILICATE (SLAG) IS PRODUCED MOLTEN SLAG IS RUN OFF AND COOLED E CaO + SiO₂ CaSiO₃ Now move the cursor away from the tower CaCO₃ CaO + CO₂
19. **THE BLAST FURNACE MOLTEN** IRON RUNS TO THE BOTTOM OF THE FURNACE. IT IS TAKEN OUT (CAST) AT REGULAR INTERVALS F CAST IRON - cheap and easily moulded - used for drainpipes, engine blocks Now move the cursor away from the tower
20. **THE BLAST FURNACE HOT WASTE GASES** ARE RECYCLED TO AVOID POLLUTION AND SAVE ENERGY G CARBON MONOXIDE - POISONOUS SULPHUR DIOXIDE - ACIDIC RAIN CARBON DIOXIDE - GREENHOUSE GAS RECAP
21. **SLAG PRODUCTION • silica** (sand) is found with the iron ore • it is removed by reacting it with limestone • calcium silicate (SLAG) is produced • molten slag is run off and cooled • it is used for building blocks and road foundations
22. **SLAG PRODUCTION • silica** (sand) is found with the iron ore • it is removed by reacting it with limestone • calcium silicate (SLAG) is produced • molten slag is run off and cooled • it is used for building blocks and road foundations EQUATIONS limestone decomposes on heating CaCO₃ → CaO + CO₂ calcium oxide combines with silica CaO + SiO₂ → CaSiO₃ overall CaCO₃ + SiO₂ → CaSiO₃ + CO₂
23. **WASTE GASES AND POLLUTION** SULPHUR DIOXIDE • sulphur is found in the coke; sulphides occur in the iron ore • burning sulphur and sulphides S + O₂ → SO₂ produces sulphur dioxide • sulphur dioxide gives SO₂ + H₂O → H₂SO₃ rise to acid rain sulphurous acid CARBON DIOXIDE • burning fossil fuels increases the amount of this greenhouse gas
24. **LIMITATIONS OF CARBON REDUCTION** Theoretically, several other important metals can be extracted this way but are not because they combine with the carbon to form a carbide e.g. Molybdenum, Titanium, Vanadium, Tungsten
25. **STEEL MAKING Iron produced** in the blast furnace is very brittle due to the high amount of carbon it contains. In the Basic Oxygen Process, the excess carbon is burnt off in a converter and the correct amount of carbon added to make steel. Other metals (e.g. chromium) can be added to make specialist steels. Removal of impurities SILICA add calcium oxide CaO + SiO₂ → CaSiO₃ CARBON add oxygen C + O₂ → CO₂ PHOSPHORUS add oxygen 2P + 5O₂ → P₄O₁₀ SULPHUR add magnesium Mg + S → MgS
26. **TYPES OF STEEL MILD** easily pressed into shape chains and pylons LOW CARBON soft, easily shaped HIGH CARBON strong but brittle chisels, razor blades, saws STAINLESS hard, resistant to corrosion tools, sinks, cutlery (contains chromium and nickel) COBALT can take a sharp edge high speed cutting tools can be magnetised permanent magnets MANGANESE increased strength points in railway tracks NICKEL resists heat and acids industrial plant, cutlery TUNGSTEN stays hard at high temps high speed cutting tools
27. **TITANIUM**
28. **EXTRACTION OF TITANIUM •** titanium ores (titanium(IV) oxide - TiO₂) are very common • titanium however is not used extensively as its extraction is difficult using conventional methods • the oxide can be reduced by carbon but the titanium produced reacts with the carbon to give titanium carbide • the extraction is a batch process so there is much time wasted and heat lost; this makes it even more expensive
29. **EXTRACTION OF TITANIUM •** the oxide is first converted to the chloride TiO₂(s) + 2C(s) + 2Cl₂(g) → TiCl₄(l) + 2CO(g) • which is then reduced with sodium. TiCl₄(l) + 4Na(s) → Ti(s) + 4NaCl(s) The reduction of TiCl₄ is carried out in an atmosphere of argon because the titanium reacts with oxygen at high temperatures.
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31. **ALUMINIUM**
32. **EXTRACTION OF ALUMINIUM Aluminium** is above carbon in the series so it cannot be extracted from its ores in the same way as carbon. Electrolysis of molten aluminium ore (alumina) must be used As energy is required to melt the alumina and electrolyse it, a large amount of energy is required.
33. **EXTRACTION OF ALUMINIUM RAW MATERIALS BAUXITE** aluminium ore Bauxite contains alumina (Al₂O₃ aluminium oxide) plus impurities such as iron oxide – it is purified before use.
34. **EXTRACTION OF ALUMINIUM RAW MATERIALS BAUXITE** aluminium ore Bauxite contains alumina (Al₂O₃ aluminium oxide) plus impurities such as iron oxide – it is purified before use. CRYOLITE Aluminium oxide has a very high melting point. Adding cryolite lowers the melting point and saves energy.
35. **EXTRACTION OF ALUMINIUM ELECTROLYSIS Unlike** iron, aluminium cannot be extracted using carbon. (Aluminium is above carbon in the reactivity series)
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37. **EXTRACTION OF ALUMINIUM ELECTROLYSIS Unlike** iron, aluminium cannot be extracted using carbon. (Aluminium is above carbon in the reactivity series) Reactive metals are extracted using electrolysis Electrolysis is expensive - it requires a lot of energy... - ore must be molten (have high melting points) - electricity is needed for the electrolysis process
38. **EXTRACTION OF ALUMINIUM ELECTROLYSIS SOLID** IONIC COMPOUNDS DON'T CONDUCT ELECTRICITY THIS IS BECAUSE THE IONS ARE NOT FREE TO MOVE
39. **EXTRACTION OF ALUMINIUM ELECTROLYSIS SOLID** IONIC COMPOUNDS DON'T CONDUCT ELECTRICITY THIS IS BECAUSE THE IONS ARE NOT FREE TO MOVE DISSOLVING IN WATER or... MELTING ALLOWS THE IONS TO MOVE FREELY
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41. **EXTRACTION OF ALUMINIUM**

42. **EXTRACTION OF ALUMINIUM CARBON** ANODE THE CELL CONSISTS OF A CARBON ANODE
43. **EXTRACTION OF ALUMINIUM STEEL CATHODE CARBON LINING THE** CELL CONSISTS OF A CARBON LINED STEEL CATHODE
44. **EXTRACTION OF ALUMINIUM MOLTEN ALUMINA** and CRYOLITE ALUMINA IS DISSOLVED IN MOLTEN CRYOLITE Na3AlF6 SAVES ENERGY - the mixture melts at a lower temperature
45. **EXTRACTION OF ALUMINIUM MOLTEN ALUMINA** and CRYOLITE ALUMINA IS DISSOLVED IN MOLTEN CRYOLITE Na3AlF6 aluminium and oxide ions are now free to move
46. **EXTRACTION OF ALUMINIUM POSITIVE ALUMINIUM** IONS ARE ATTRACTED TO THE NEGATIVE CATHODE $Al^{3+} + 3e^- \rightarrow Al$ EACH ION PICKS UP 3 ELECTRONS AND IS DISCHARGED CARBON CATHODE
47. **O²⁻ + 2e⁻** EXTRACTION OF ALUMINIUM NEGATIVE OXIDE IONS ARE ATTRACTED TO THE POSITIVE ANODE EACH ION GIVES UP 2 ELECTRONS AND IS DISCHARGED CARBON ANODE
48. **EXTRACTION OF ALUMINIUM ELECTRONS CARBON** ANODE CARBON CATHODE
49. **EXTRACTION OF ALUMINIUM ELECTRONS ANODE** $3O^{2-} - 1\frac{1}{2}O_2 + 6e^-$ OXIDATION OXIDATION (LOSS OF ELECTRONS) TAKES PLACE AT THE ANODE CARBON ANODE
50. **EXTRACTION OF ALUMINIUM ELECTRONS CATHODE** $2Al^{3+} + 6e^- \rightarrow 2Al$ REDUCTION OXIDATION (LOSS OF ELECTRONS) TAKES PLACE AT THE ANODE REDUCTION (GAIN OF ELECTRONS) TAKES PLACE AT THE CATHODE CARBON CATHODE ANODE $3O^{2-} - 1\frac{1}{2}O_2 + 6e^-$ OXIDATION
51. **EXTRACTION OF ALUMINIUM ELECTRONS OXIDATION** (LOSS OF ELECTRONS) TAKES PLACE AT THE ANODE REDUCTION (GAIN OF ELECTRONS) TAKES PLACE AT THE CATHODE CARBON ANODE CARBON CATHODE ANODE $3O^{2-} - 1\frac{1}{2}O_2 + 6e^-$ OXIDATION CATHODE $2Al^{3+} + 6e^- \rightarrow 2Al$ REDUCTION
52. **EXTRACTION OF ALUMINIUM CARBON** ANODE PROBLEM THE CARBON ANODES REACT WITH THE OXYGEN TO PRODUCE CARBON DIOXIDE CARBON DIOXIDE
53. **EXTRACTION OF ALUMINIUM CARBON** ANODE PROBLEM THE CARBON ANODES REACT WITH THE OXYGEN TO PRODUCE CARBON DIOXIDE THE ANODES HAVE TO BE REPLACED AT REGULAR INTERVALS, THUS ADDING TO THE COST OF THE EXTRACTION PROCESS CARBON DIOXIDE
54. **PROPERTIES OF ALUMINIUM ALUMINIUM** IS NOT AS REACTIVE AS ITS POSITION IN THE REACTIVITY SERIES SUGGESTS THIS IS BECAUSE A THIN LAYER OF ALUMINIUM OXIDE QUICKLY FORMS ON ITS SURFACE AND PREVENTS FURTHER REACTION TAKING PLACE THIN LAYER OF OXIDE ANODISING PUTS ON A CONTROLLED LAYER SO THAT THE METAL CAN BE USED FOR HOUSEHOLD ITEMS SUCH AS PANS AND ELECTRICAL GOODS
55. **CHROMIUM**
56. **EXTRACTION OF CHROMIUM The** method of extraction often depends on the purity required. IMPURE CHROMIUM The ore (chromite) is reduced by heating with carbon. ... $FeCr_2O_4(s) + 4C(s) \longrightarrow Fe(s) + 2Cr(s) + 4CO(g)$
57. **EXTRACTION OF CHROMIUM The** method of extraction often depends on the purity required. IMPURE CHROMIUM The ore (chromite) is reduced by heating with carbon. ... $FeCr_2O_4(s) + 4C(s) \longrightarrow Fe(s) + 2Cr(s) + 4CO(g)$ PURE CHROMIUM The chromite is converted to chromium(III) oxide which is then reduced using aluminium at high temperatures. This is known as ACTIVE METAL REDUCTION. $Cr_2O_3(s) + 2Al(s) \longrightarrow 2Cr(s) + Al_2O_3(s)$
58. **SODIUM**
59. **EXTRACTION OF SODIUM Involves** electrolysis of molten sodium chloride in the Down's Cell. CaCl₂ is mixed with the sodium chloride to lower the melting point and reduce energy costs. Sodium is discharged at the cathode $Na^+ + e^- \longrightarrow Na$ Chlorine is discharged at the anode $Cl^- \longrightarrow \frac{1}{2}Cl_2 + e^-$
60. **RECYCLING Problems** • high cost of collection and sorting • unsightly plant • high energy process Social • less visible pollution of environment by waste benefits • provides employment • reduces the amount of new mining required Economic • maintains the use of valuable resources benefits • strategic resources can be left underground
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